Design and Optimization of a Fast-Switching Electromagnetic Valve Actuator for Off-highway Vehicle Hydraulic Systems

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***Abstract*—Electrification of off-highway vehicles offers improved efficiency and reduction in greenhouse gas emissions. However, the extremely low power densities of electric machines relative to the hydraulics used in conventional off-highway vehicles makes electrification challenging. Although the recently proposed Hybrid Hydraulic Electric Architecture (HHEA) can overcome this limitation, it requires frequent switching between the pressure rails using switching valves resulting in losses. Fast switching minimizes the switching losses in these valves, which otherwise significantly reduce the overall efficiency of the system. The primary contribution of this paper is the design and optimization of a linear electromagnetic actuator for a fast-switching valve. This paper identifies the actuator requirements to enable fast switching and presents a design and optimization technique to realize actuators that meet the design objectives. In addition, a multi-physics approach is proposed to model the performance of the complete switching valve by considering the electromagnetic, fluidic, and mechanical physics. An optimal actuator design is obtained using the developed optimization framework, and the multi-physics model demonstrates that the selected design achieves a 30% reduction in throttling-related switching losses by switching 70% faster than comparable commercial valve designs.**

***Index Terms*—Linear actuators, off-highway vehicle electrification, multi-physics simulation, optimization, electromagnetic design**

I. Introduction

Off-highway vehicles are mobile machines that are typically used for performing specific tasks like excavation or digging in applications such as agriculture, construction, and forestry. In conventional off-highway vehicles, these tasks are performed by implements that utilize hydraulic actuators. However, the throttle based control of these implements yields efficiencies as low as 21% from the engine shaft to the implement [1]. Moreover, since conventional vehicles are powered by diesel engines, poor system efficiency leads to added carbon emissions. The goal of reducing environmental impact and promoting sustainable energy solutions is propelling the shift towards the electrification of off-highway vehicles. The electrification of off-highway vehicle implements such as the one shown in Fig. 1a provides benefits like enhanced efficiency, improved control, the ability to regenerate energy, and reduced greenhouse gas emissions [2]. The significantly lower torque densities of electric machines make it difficult to directly substitute them for inefficient hydraulic actuators [3], [4]. However, the implementation of hybrid systems that combine electric and hydraulic components offers a solution to overcome the power density limitations of electric machines. One such architecture is the Hybrid Hydraulic Electric Architecture (HHEA) proposed in [3] and shown in Fig. 1a.

The HHEA combines the power density advantage of hydraulics with the efficiency and controllability benefits of electric machines by having a set of common pressure rails (CPR) and an electric motor-driven hydraulic pump to control the implements. More details on the functionality and benefits of the HHEA can be found in [3].

Efficient switching is critical to the HHEA due to the requirement of frequent switching between the CPR. Each switching event involves closing the on-off valve for the current pressure rail and opening the valve for the next selected pressure rail, resulting in switching losses due to fluid compressibility and throttling. The aim of this paper is to address the throttling losses that occur during the finite transition period of switching events, where the partially open valve causes energy loss due to throttling across the valve. Presently, switching losses make up around 20-25% of the total hydraulic losses in the HHEA [6], highlighting the importance of reducing these losses to enhance the architecture's efficiency.

One potential method of minimizing these losses is to increase the speed of valve switching [7]. By doing so, the transition period between fully open and fully closed states can be significantly reduced, which in turn minimizes the throttling losses. The switching valves typically comprises a moving plunger that an actuator (either electromagnetic or hydraulic) controls to switch the flow. The objective of this paper is to design the actuator for a direct-acting switching valve that enables fast switching and reduces throttling-related switching losses.

This paper characterizes the design space of the electromagnetic actuators for achieving fast switching (< 13 ms switching time) to reduce switching losses in hydraulic valves carrying

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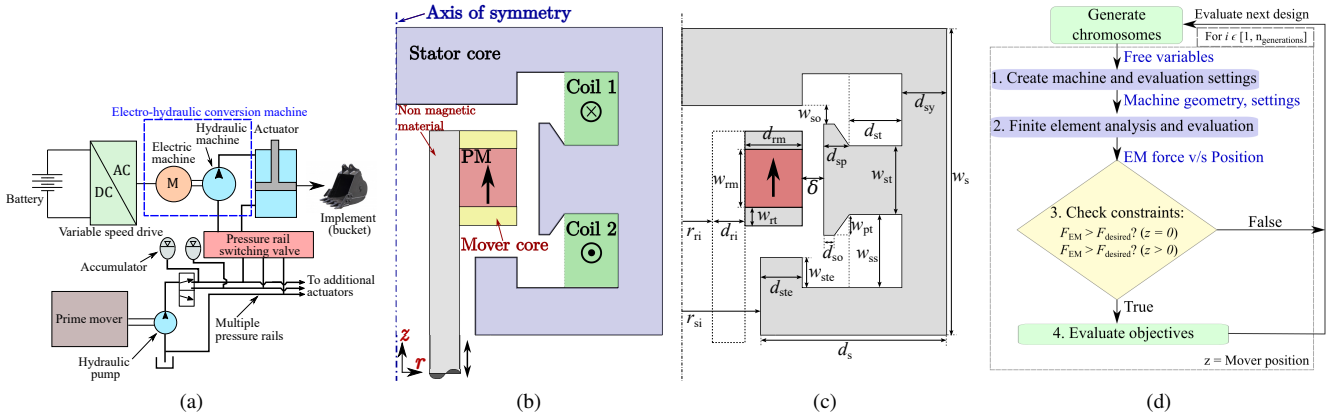


Fig. 1. (a) Schematic of the HHEA architecture for one implement [5]; (b) Selected bistable electromagnetic actuator topology; (c) Parameterized design; (d) Optimization algorithm flowchart

large flow rates. The primary contributions of this paper are: 1) actuator design to minimize valve switching losses, 2) development of a multi-objective optimization framework for fast switching valve actuators, and 3) development of a multi-physics modeling framework to evaluate the valve's switching performance. Although the valve actuator requirements in this paper are derived specifically for the HHEA, the design methodology and findings presented will benefit all applications requiring fast-switching valves, such as digital hydraulics. This paper is organized as follows: Section II presents the state-of-the-art in fast-switching valve actuation and derives the actuator requirements, analytical design of the actuator, and the design optimization are presented in Sections III and IV, finally, the multi-physics modeling approach to simulate valve performance is discussed in Section V.

II. Fast Switching Valve Actuators

This section establishes the actuator requirements for fast-switching valves in the HHEA and reviews the state-of-the-art in fast-switching valve actuator mechanisms and designs. This study considers the electrification of a compact track loader using a 3-rail HHEA as shown in Fig. 1a for determining the valve actuator requirements. The system specifications and requirements are listed in Table I.

A. Valve actuator requirements

The switching valve in the HHEA needs to operate for a maximum pressure difference of the highest pressure rails for the architecture. For heavy duty mobile machines, the highest pressure rail is chosen to be 35 MPa. The torque or force rating of the hydraulic actuator for the implement governs the flow rate rating for the switching valve. For the application under study, the valve needs to operate at a maximum flow of 45 L/min and a 1 MPa pressure drop for the maximum flow when the valve is fully open. To minimize leakage and ensure fast response times, a direct-acting poppet valve design has been chosen for the switching valves. Poppet valves are known for their simple construction, fast response time, and ability to handle high flow rates [8].

TABLE I
System specification and valve actuator requirements

Specification	Rating	Units
No. of CPR	3	
No. of fast switching valves	6	
Max. input pressure	35	MPa
Max. flow rate	45	L/min.
Valve open state pressure drop	1	MPa
Max. switching time	13	ms
Min. actuator force	90	N
Min. stroke length	2	mm

To derive the switching time and the actuator force requirements, a 45 L/min. poppet valve design shown in Fig. 2a is evaluated based on the model presented in Section V to determine the hydraulic switching energy loss and actuator force requirement for a range of switching times and the results are shown in Fig. 2b. The results show that reducing the switching time below 13 ms only minimally reduces throttling losses. As a result, the potential benefits of switching faster than 13 ms are outweighed by the added costs and complexity involved in achieving faster switching times. As seen in Fig. 2b, the actuator needs to generate approximately 90 N of force to achieve 13 ms switching time. It is to be noted that Fig. 2b shows a constant step actuator force requirement. In a practical scenario, the force generated by the actuator needs to be higher than 90 N to account for the delay in force build-up and the mover mass acceleration. Based on the valve flow rate of 45 L/min. and fully open state pressure drop requirement of 1 MPa, a stroke length of 2 mm is required for a cone angle α of 120 degrees for the poppet valve.

B. State of the art

Valves can be actuated by either hydraulic or electromagnetic actuators. For a direct-driven poppet valve architecture, electromagnetic actuation enables leak-proof operation, high power density, high actuation force, and high efficiency [9], all of which are of prime importance for off-highway vehicles. Hence, electromagnetic actuation is considered in this study. Valve designs that use other force generating mechanisms like piezoelectric actuators, magnetostrictive actuators, and active

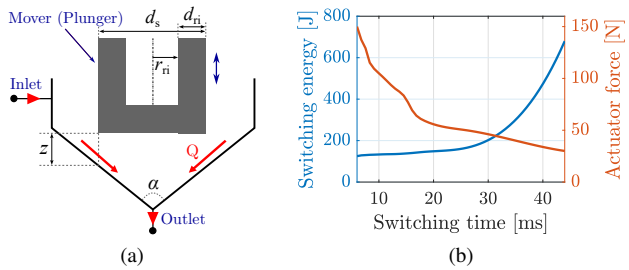


Fig. 2. (a) Poppet valve geometry; (b) Effect of valve's switching time on fluidic switching energy loss and electromagnetic actuator force requirement

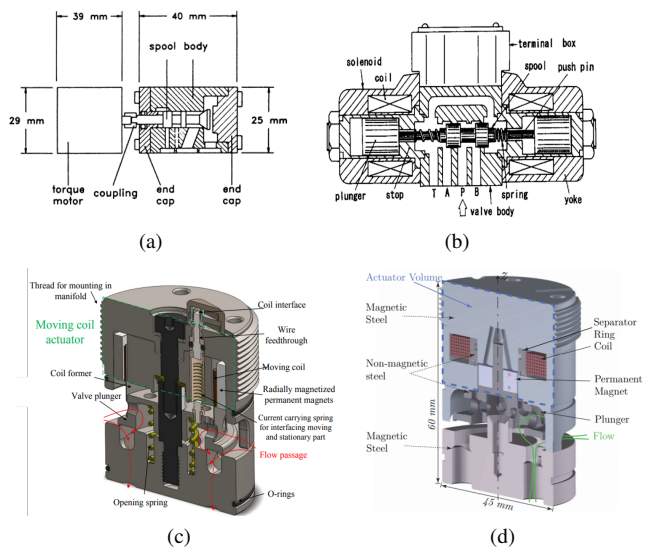


Fig. 3. Common types of hydraulic valve actuators: (a) Rotary actuator [12] (b) Linear reluctance actuator [13] (c) Linear moving coil actuator [14]; (d) Linear moving magnet actuator [15]

material based actuation exist in literature but these designs can only have small stroke lengths and hence they are not suitable for actuating hydraulic valves [10], [11]. Figure 3 shows some common valve designs that are reviewed.

Another design choice for the actuator is the motion of actuation which may be linear or rotary. A linear actuator is considered in this study since rotary designs can result in complex valve structures, e.g.: [10], [16] that are not well-suited for applications with varying duty cycles of on and off time, such as rail switching in off-highway vehicles. Linear actuators may be classified into two categories based on the force generating mechanism: Reluctance force and Lorentz force. The actuators using Lorentz force typically use a permanent magnet either on the mover (moving magnet actuators) or on the stator (moving coil actuators). Typically, the reluctance force actuators are slower as compared to the moving magnet or moving coil actuators [17]. This is because the coil current takes time to build up the flux in the core that produces the force. The moving coil actuators have a lower mover mass due to the presence of the coils as opposed to permanent magnets on the mover and this results in a faster transient response. However, a connection has to be made with the moving coil as shown in Fig. 3c which impacts the robustness of the actuator.

The linear actuators can also be categorized based on the mode of operation as monostable and bistable. Monostable designs generally utilize spring force or fluid force to hold the mover at one end (either open or close) and use electromagnetic force to hold it at the other end. Naturally, monostable designs consume energy both during the switching transition and during the state when electromagnetic force is holding the mover to one end. As identified in [9], bistable actuators consume energy only during a switching transition, have lower average current density, and can have zero power latching making them more efficient.

Commercially available valves with a significant flow rating either have high switching time or use pilot actuation which is not desirable for HHEA application. Many commercial designs implement a monostable actuation which results in high electrical losses and heat generation in an application like HHEA [11], [18]. For this project, a bistable two slot, one pole-pair, single phase tubular linear machine with a ring-shaped permanent magnet mounted on a non-magnetic material as shown in Fig. 1b is selected. This topology is preferred because of its simple geometry, tubular design that results in low coil resistance and inductance, bistable operation and high latching force. While a similar design has been investigated in [9], [15] and [19] for different applications such as digital displacement machines, there is no published literature that details the analytical electromagnetic design, multi-objective electromagnetic optimization and multi-physics model based simulation for valve actuators that this paper develops.

III. Analytical Electromagnetic Design

In this section, the analytical design for the actuator is presented for the tubular design shown in Fig. 1b. A fully parameterized version of the design to support the analytic design and electromagnetic design optimization is shown in Fig. 1c. In such a machine, a square wave current excitation of the coils results in a square wave force profile on the mover. For a desired peak electric loading $\hat{A}$, peak magnetic loading $\hat{B}_\delta$, and effective air gap area A_δ, the electromagnetic force generated by the mover can be computed as $F_{em} = \frac{1}{\sqrt{2}} \cdot \hat{A} \cdot \hat{B}_\delta \cdot A_\delta$ [20, Sec. II-B]. The stator core dimensions are determined based on the maximum desired flux density in each region. Other key dimensions are derived as shown in (1); where, Q is the number of slots, p is the number of pole pairs and S_u is the physical slot area available for the coil. Outer dimensions, w_s and d_s determine the size of the actuator and the inner radius dimension, r_{ri} can be used to vary the equivalent air gap area, A_δ for achieving the desired force.

$$w_{\mathrm{st}} = \frac{w_{\mathrm{s}}}{Q}\frac{\hat{B}_\delta}{B_{\mathrm{st}}}; \quad d_{\mathrm{sy}} = \frac{w_{\mathrm{s}} \cdot \hat{B}_\delta}{2pB_{\mathrm{sy}}}; \quad w_{\mathrm{ss}} = \frac{w_{\mathrm{s}} - (Q \cdot w_{\mathrm{st}})}{Q}$$

$$d_{\mathrm{st}} = \frac{S_{\mathrm{u}}}{w_{\mathrm{ss}}}; \quad d_{\mathrm{rm}} = \frac{\mu_{\mathrm{rm}}\delta\hat{B}_\delta}{B_{\mathrm{r}} - \hat{B}_\delta} \tag{1}$$

The remaining dimensions are derived as a ratio or as a fraction of these key dimensions. For example, d_{ste} is set to be equal to d_{sy} as both regions carry the same flux and d_{sy} is set to be 70 % of w_{st} since the flux carried by the stator yoke is about half the flux carried by the tooth.

TABLE II
OPTIMIZATION FREE VARIABLES AND DIMENSIONS FOR THE SELECTED DESIGN

Free variable	Symbol	Bounds [mm]	Selected design [mm]
Stator tooth width	w_{st}	[3.1, 7.6]	3.1
Stator pole thickness	d_{sp}	[1.4, 3.0]	2.5
Stator tooth depth	d_{st}	[4.5, 17.1]	16.3
Slot height	w_{ss}	[8.7, 23.7]	10.9
Slot opening	w_{so}	[0.7, 7.0]	3.8
Airgap length	δ	[0.62, 2.5]	0.64
Permanent magnet thickness	d_{rm}	[3.0, 9.0]	8.9
Mover iron thickness	w_{rt}	[2.4, 5.1]	2.4
Permanent magnet height	w_{rm}	[3.5, 11.9]	3.7

IV. ELECTROMAGNETIC DESIGN OPTIMIZATION

The performance of the design from Fig. 1c is sensitive to design dimensions. For example, too small of a slot opening w_{so} results in large leakage flux, and too large of a slot opening reduces the equivalent air gap area A_δ, resulting in lower electromagnetic force. Similarly, the mover acceleration and subsequently the switching time depends on the mover mass which in turn depends on the mover core and the permanent magnet geometry. This necessitates exploring the design space using an optimization framework.

A. Optimization study

Figure 1d shows the flowchart representation for the optimization setup developed to explore the design space. The optimization algorithm generates a population of designs, performs FEA (Finite Element Analysis) evaluation for each design to evaluate the objectives and check for constraints. The designs violating the constraints (minimum latching force LF and minimum moving force MF) are discarded. A constraint is imposed on the minimum latching force to ensure that the mover remains in its stable state of fully open or fully closed, even when subjected to disturbances caused by fluid forces. The minimum moving force requirement is derived from the analysis presented in section II. The optimization framework is implemented in Python and uses an open-source FEA tool to perform a static evaluation for each design. Table II lists the free variables and their bounds used for the optimization study.

The objective of the optimization study is to search for a small and compact design that generates the required force to meet the switching time criteria. Minimizing the mover mass MM helps in achieving higher mover acceleration for a given force and results in lower switching time. The force ripple FR affects the force generated during the start of the motion. It is desired to have a design with low force ripple. The size of a design directly affects the active material cost AMC and it is desirable to have low cost. It is assumed that the steel core has a cost of 17087 $\$/\mathrm{m}^3$, the permanent magnet has a cost of 708400 $\$/\mathrm{m}^3$, and the cost of copper is 73222 $\$/\mathrm{m}^3$. The objective and constraint functions used for the optimization are summarized in (2) and (3).

$$O_1 : \min(\mathrm{FR})\,;\ O_2 : \min(\mathrm{MM})\,;\ O_3 : \min(\mathrm{AMC}) \tag{2}$$

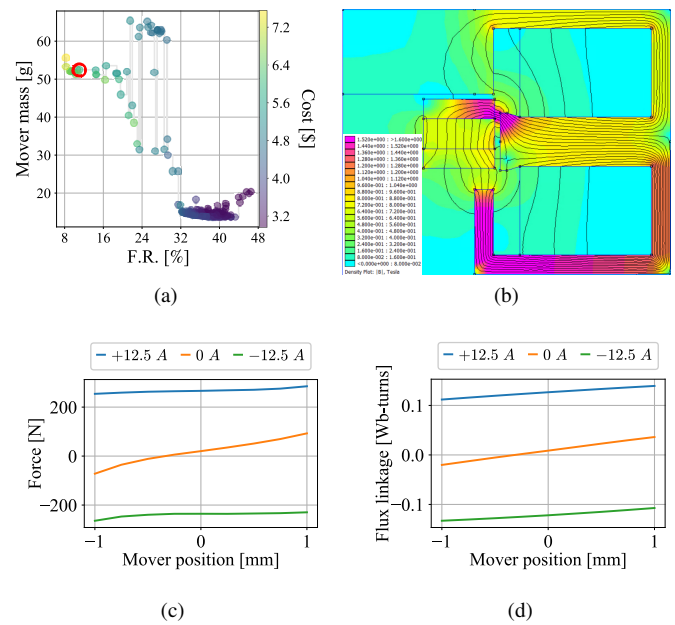


Fig. 4. (a) Pareto front after 35 generations; Selected design: (b) Cross-sectional flux density map; (c) Force profile; (d) Flux profile

$$C_1 : \mathrm{LF} > 18\ \mathrm{N};\quad C_2 : \mathrm{MF} > 100\ \mathrm{N} \tag{3}$$

B. Optimization results

Figure 4a shows the Pareto front after evaluating more than 5000 designs. The general trend is that the designs with high force ripple have low cost and vice versa. Similarly, designs with high mover mass have lower force ripple.

A design from the pareto front having 267 N of average force that meets the performance requirements is selected (marked with a red circle in Fig. 4a). The free variable values for the design are listed in Table II. The cross-sectional flux density contour plot, and force and flux linkage profiles for this design are shown in Fig. 4b, Fig. 4c and Fig. 4d. The height (w_{s}) of the selected design is 34 mm and the radius ($r_{\mathrm{si}} + d_{\mathrm{s}}$) is 40.8 mm. This particular design is selected for further analysis because of its compactness and low force ripple of 10.9%. Furthermore, the design has 0.86 Ω resistance and 9.5 mH inductance which results in a smaller time constant of 11.04 ms as compared to the median time constant of 12.2 ms for the designs in the pareto front with force ripple less than 25%. The low time constant helps in faster current and subsequently faster force build up when the coil is excited. This results in smaller switching time.

V. MULTI-PHYSICS MODELING

A switching valve is a multi-physics system consisting of fluidic, mechanical, and electromagnetic domains as shown in Fig. 5 [11]. This section proposes a multi-physics modeling approach to simulate fast-switching valve performance. The proposed modeling approach allows for verifying the actuator design performance in the switching valve application by evaluating the complex interactions between the electromagnetic, mechanical, and fluid physics. The proposed simulation model

is used to identify a suitable design out of a set of candidate designs by performing a detailed multi-physics simulation.

A. Mechanical domain

The mechanical domain of the multi-physics model is the force balance equation, which is used to determine the mover acceleration based on the electromagnetic and fluidic forces acting on it. Ignoring the gravitational force on the mover and the Stribeck friction force, the force balance equation governing the mover motion for a bistable actuator is given by (4) [21], where, m is the mover mass, z is the mover position, F_{em} is the electromagnetic force generated by the actuator, and F_{fluid} is the fluidic force on the mover.

$$m\ddot{z} = F_{\mathrm{em}}(z) + F_{\mathrm{fluid}}(z, Q, \Delta P, t) \tag{4}$$

B. Fluidic domain

The fluid domain of the multi-physics model constitutes the flow forces and the viscous damping on the mover (5). The fluid forces prevent the opening of the poppet valve shown in Fig. 2a and they need to be overcome by the electromagnetic force. The viscous damping force is a result of the moving plunger inside the fluid and is proportional to the velocity of the plunger. This force is given by (6), where; m is the mass of the plunger, $\dot{z}$ is the velocity of the plunger and c is the damping constant assumed to be 4000.

$$F_{\mathrm{fluid}} = F_{\mathrm{flow}}(z, Q, \Delta P) + F_{\mathrm{damp}}(\dot{z}) \tag{5}$$

$$F_{\mathrm{damp}} = mc\dot{z} \tag{6}$$

Only the steady-state flow forces are considered in this study. The flow forces are a result of the flowing fluid inside the valve during the switching interval. The orifice equation (8) is used to calculate the flow Q across the valve and the resulting fluid force is given by (7); where, ρ is the density of the hydraulic fluid, ΔP is the pressure difference across the valve, d_{s} is the plunger diameter, α is the cone angle of the poppet valve, A_{s} is the area of opening available for the fluid to flow (represented by (10) as a function of the stroke length z), and θ is the jet angle given by the empirical relation (9) [22].

$$F_{\mathrm{flow}} = \frac{\rho\, Q^2 \cos(\theta)}{A_{\mathrm{s}}} \tag{7}$$

$$Q = C_{\mathrm{d}} A_{\mathrm{s}} \sqrt{\frac{2\Delta P}{\rho}} \tag{8}$$

$$\theta = 0.3663 + 0.8373\ \left(1 - \exp\frac{-z}{1.848}\right) \tag{9}$$

$$A_{\mathrm{s}} = \left(d_{\mathrm{s}} + z\ \sin\frac{\alpha}{2}\ \cos\frac{\alpha}{2}\right)\ z\ \sin\frac{\alpha}{2} \tag{10}$$

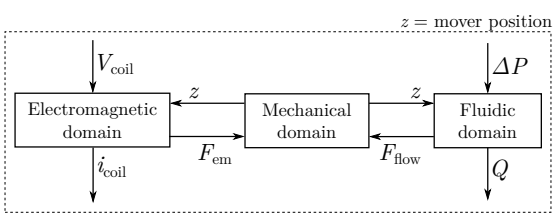


Fig. 5. Block diagram representation of the multi-domain valve system

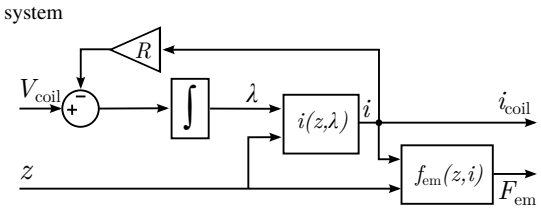


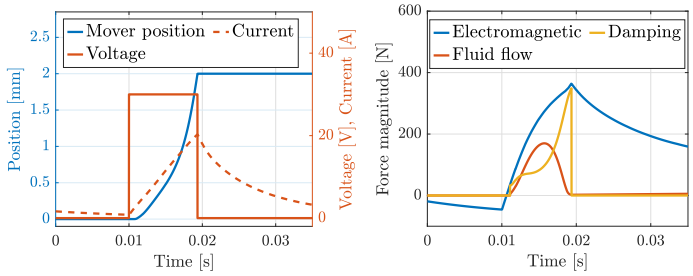
Fig. 6. Block diagram representation of the electrical domain

C. Electromagnetic domain

The Lorentz force resulting from the interaction between the permanent magnet field and the magnetic field generated by the coils is the primary force generating mechanism for the selected actuator topology shown in Fig. 1b. The resulting electromagnetic force is a non-linear function of the coil current and the mover position. Ideally, the model needs to perform FEA to evaluate the electromagnetic force for every coil current value and the mover position to accurately represent the electromagnetic domain of the actuator. However, this is computationally expensive. A faster approach is to perform the FEA of the selected design at a range of coil currents and a range of mover positions in the air gap. Resulting force profile and flux linkage profiles similar to the ones shown in Figs. 4c, 4d for each coil current are recorded. Then, the block diagram shown in Fig. 6 is used to obtain the electromagnetic force value for each operating condition as determined by the flux linkage (volt-seconds) and the mover position. The flux linkage profiles obtained from FEA are inverted and fed into a lookup table to obtain the operating coil current for any instantaneous flux linkage and mover position values. This task is relatively simpler for obtaining the electromagnetic force. The force profile data for a range of coil currents, typically from 0 to about 1.5 times the rated coil current, is directly fed into the model as a lookup table which is used to obtain the instantaneous electromagnetic force for any position and coil current value.

D. Multi-physics evaluation results

A step positive voltage command is applied to open the valve and a step negative voltage command closes the valve. As discussed in section II, the selected bistable valve topology allows for the removal of the supply voltage after the transition is complete. For simplicity, the duration of the voltage command pulse is kept constant at 9 ms. The multi-physics simulation results for a 2 mm stroke length for the mover from a close position to an open position; and the coil current and supply voltage are shown in Fig. 7a. The electrical input energy for one switching transition is 2.7 J. The forces acting on the mover during the transition are shown in Fig. 7b. The fluid flow force and the damping force are represented in the positive direction in Fig. 7b, however in reality, these forces oppose the electromagnetic force and reduce the effective acceleration of the mover as per (4). The magnitude of the



(a) (b)

Fig. 7. Multi-physics simulation results for a valve opening transition: (a) Mover position, coil current, and coil voltage; (b) Forces acting on the mover

stiff voltage source governs the coil current and determines the valve switching time. More sophisticated control for the selected valve topology may include force control or position control using current sensors and position feedback.

The chosen valve design is able to switch within 9 ms, resulting in approximately 132 J of switching energy loss due to throttling. Considering 5 Hz switching frequency for the HHEA, the throttling-related switching loss with the proposed design is 660 W against 1070 W for a commercially available direct-acting poppet valve having 30 ms switching time for a comparable flow rating [11].

VI. Conclusion

This paper presents a procedure for designing a linear electromagnetic actuator for fast-switching hydraulic valves to reduce throttling-related switching loss in applications like the HHEA and digital hydraulics that require frequent switching. First, the literature on fast-switching valve actuators is reviewed to identify a tubular linear bistable topology for analysis. Next, an analytical design procedure is developed to obtain an initial electromagnetic design based on the actuator force requirement. An optimization framework is developed and used to explore the design space for the actuator by investigating the trade-off between practically relevant metrics. The results show that the designs have a trade-off relation between the force ripple and cost. An optimal design having a low force ripple of 10.9% and a small coil inductance is selected for further analysis. Finally, a multi-physics modeling approach is developed to validate the actuator performance. This framework is used to show that the selected optimal design achieves a 9 ms switching time and reduces the throttling-related switching losses by 38% compared to a commercial valve with approximately 30 ms switching time. These findings demonstrate the potential of the developed fast-switching valve actuator to improve the efficiency of the HHEA system and motivate future work to fabricate the optimal design and experimentally validate its performance.

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